Predicting Hotel Booking Cancellations: a CRISP-DM Approach

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***Abstract*—Booking cancellations are a major source of revenue loss for hotels, since an unsold room-night cannot be recovered. This work follows the CRISP-DM methodology to predict whether a booking will be cancelled using only the information available at booking time. After removing duplicated records and attributes that leak the outcome of the booking, four classifiers (Logistic Regression, Decision Tree, Random Forest and XGBoost) are compared using metrics suited to an imbalanced target. XGBoost achieves the best PR-AUC (0.778) on a held-out test set and, with a decision threshold of 0.4, detects 89% of the cancellations. A risk-based segmentation shows that 24% of the bookings concentrate 63% of the cancellations, supporting targeted deposit and reminder policies.**

1. Introduction

In the hotel industry, a room is a perishable product: if it stays empty for one night, that revenue is lost forever. Cancellations are therefore a central problem for revenue management. Their impact depends on *when* they happen: a booking cancelled months in advance leaves time to resell the room, while a last-minute cancellation is much harder to recover. Hotels typically react with restrictive policies, such as non-refundable rates and deposits, or with overbooking, which can however damage customer satisfaction when too many guests show up.

The goal of this project is to predict whether a booking will be cancelled (`is_canceled`) and to identify the bookings and customer profiles most at risk, so that the hotel can apply targeted actions instead of the same policy for every guest. The analysis uses the *Hotel Booking Demand* dataset [1], which contains real bookings of a City Hotel and a Resort Hotel in Portugal between 2015 and 2017, and follows the CRISP-DM methodology [2]. The code is available as a Google Colab notebook in the project repository.

2. Methodology

2.1. Business Understanding

The model is intended to support two low-cost actions: *contacting* high-risk guests before arrival and *requiring a deposit* for the riskiest bookings. In this scenario a false negative (a cancellation that is not detected) is the most costly error, since the room may remain empty, while a false positive only causes an extra reminder or a stricter booking condition. Recall on the cancelled class is therefore the priority. Since the target is imbalanced, accuracy is misleading; Precision, F1-score and the area under the Precision-Recall curve (PR-AUC) are also used, the latter being more informative than ROC-AUC on imbalanced data [3].

2.2. Data Understanding

The original dataset contains 119,390 bookings and 32 attributes; 37% of the bookings are cancelled. The exploratory analysis showed that the cancellation rate grows steadily with the *lead time* (from 9.6% for bookings made within a week to 67.7% for those made more than a year in advance), is higher for the City Hotel (41.7% vs 27.8%) and for the Groups and online travel agency segments, and decreases with the number of special requests. Two findings were particularly relevant:

- *Non Refund paradox*: 99.4% of the bookings with a non-refundable deposit were cancelled. These bookings were almost all City Hotel, Portuguese, Groups or Offline TA/TO reservations, suggesting block bookings by tour operators rather than individual guests.
- *Hidden leakage*: no booking with a requested parking space was cancelled, and only 5.4% of bookings whose assigned room differed from the reserved one were cancelled. These attributes are mostly recorded at check-in, so they are only known for guests who actually arrived.

Data quality checks revealed 31,994 duplicated rows, 180 bookings with no guests, a negative and an extreme (5,400 EUR) daily rate, missing values in `company` (94%), `agent` (14%), `country` and `children`, and a few "Undefined" categories.

2.3. Data Preparation

All exact duplicates were removed: identical rows add no information and could appear both in the training and

TABLE 1. MAIN DATA PREPARATION STEPS

Step	Bookings	Cancelled
Original dataset	119,390	37.0%
Duplicates removed	87,396	27.5%
Anomalies removed	87,223	27.5%
Training set (80%)	69,778	27.5%
Test set (20%)	17,445	27.5%

TABLE 2. PERFORMANCE ON THE TEST SET

Model	Precision	Recall	F1	PR-AUC
Baseline	0.000	0.000	0.000	0.275
Logistic Regression	0.534	0.770	0.630	0.665
Decision Tree	0.555	0.826	0.664	0.718
Random Forest	0.629	0.796	**0.703**	0.767
XGBoost	0.612	0.821	0.701	**0.778**

in the test set, inflating the results. After this step 87,396 bookings remain and the cancellation rate drops to **27.5%**; non-refundable bookings drop from 14,587 to 1,037, confirming that the Non Refund paradox was largely caused by duplicated group bookings. Bookings with no guests, the anomalous rates and the rows with undefined segment or channel were also removed, obtaining 87,223 bookings (Table 1).

Missing values were handled according to their meaning: `children` was set to 0; a missing `country` became "Unknown"; a missing `agent` means that no agency was involved and became the category "No agent"; `company`, almost empty, was replaced by a binary flag `has_company`. Since `agent` and `country` are high-cardinality identifiers, only their 10 most frequent values were kept and the others grouped into "Other".

To use only information available at booking time, four attributes were removed as leakage:

- `reservation_status` and `reservation_status_date`, which directly describe the outcome;
- `assigned_room_type` and `required_car_parking_spaces`, mostly recorded at check-in.

For the same reason, the feature initially planned in the proposal (consistency between reserved and assigned room) was not created. The arrival year was also dropped, so that the model does not depend on the specific years in the data. Four features were engineered: `total_nights`, `total_guests`, `has_children` and `has_previous_cancellations`. Categorical attributes were one-hot encoded (76 features in total) and the data were split into a stratified training set (80%) and test set (20%).

2.4. Modeling

Four classifiers of increasing complexity were compared: Logistic Regression as a linear reference, a Decision Tree, Random Forest [4] (bagging) and XGBoost [5] (boosting), together with a majority-class baseline. Class imbalance was handled with class weights (`class_weight='balanced'` and `scale_pos_weight` for XGBoost) rather than synthetic oversampling. Hyperparameters were tuned on the training set only, with grid or random search and stratified cross-validation, selecting the configuration with the highest PR-AUC; features were standardised inside a pipeline for Logistic Regression. Model interpretation was performed with SHAP [6].

3. Results

3.1. Model comparison

Table 2 reports the performance on the test set. All models clearly outperform the baseline, whose PR-AUC (0.275) equals the share of cancellations. Performance increases with model complexity, suggesting non-linear relationships. Random Forest and XGBoost are very close: Random Forest has a slightly higher F1-score, while XGBoost achieves the highest PR-AUC and a higher Recall, and was therefore selected as the final model. Test scores are close to the cross-validation ones (e.g. PR-AUC 0.763 in cross-validation for XGBoost), indicating no overfitting. To check robustness over time, XGBoost was also retrained on the 2015–2016 bookings and tested on those of 2017 (January–August): performance decreases (PR-AUC 0.677, Recall 0.722 with threshold 0.4), and the cancellation rate rises from 25.0% to 31.9% between the two periods. Customer behaviour changes over time, so the random split gives an optimistic estimate and the model should be retrained regularly. These results are lower than those often reported for this dataset because duplicates and leakage attributes were removed, but they are more realistic for a model used at booking time.

3.2. Choice of the decision threshold

With the default threshold of 0.5, XGBoost detects 3,943 of the 4,801 cancellations in the test set (Recall 0.821), with 2,502 false alarms and 858 missed cancellations (Fig. 1).

Since missing a cancellation is the most costly error in the chosen scenario, a threshold of **0.4** is proposed: Recall rises to 0.887 while Precision decreases from 0.612 to 0.556. A riskier use of the model, such as overbooking, would instead require a higher threshold to limit false positives. Fig. 2 shows that the Precision-Recall curve stays well above the baseline at every level of Recall.

3.3. Drivers of cancellation

According to SHAP (Fig. 3), the most important features are the guest's country (Portuguese guests cancel more: 35.7% vs 23.7%), the lead time (longer lead times increase the risk), the number of special requests (which strongly

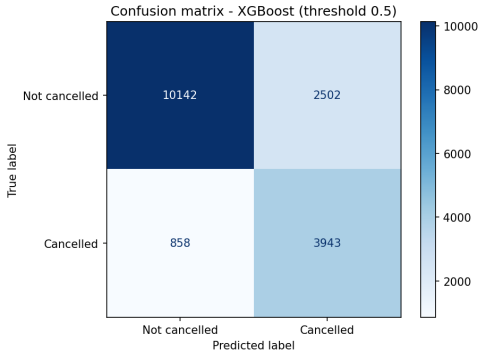


Figure 1. Confusion matrix of XGBoost on the test set (threshold 0.5).

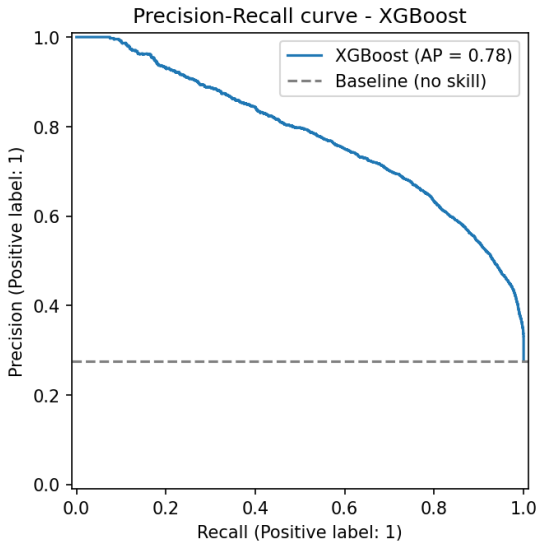


Figure 2. Precision-Recall curve of XGBoost on the test set.

reduce it, as a sign of commitment), the online travel agency segment (riskier) and the number of booking changes (associated with lower risk). After removing duplicates, the non-refundable deposit type is much less important than the exploratory analysis suggested.

3.4. Risk segmentation and recommendations

The predicted probabilities were used to divide the test bookings into three risk levels (Table 3). This risk-based segmentation was preferred to the K-Means clustering mentioned in the proposal, since it is directly linked to the hotel's decisions and avoids distance-based clustering on mostly one-hot encoded features. The high-risk group contains only 24% of the bookings but **63% of all cancellations**; typical high-risk bookings are made about four months in advance, through online agencies and with few special requests.

These results suggest a differentiated policy: a partial deposit or stricter cancellation conditions for high-risk bookings (rather than fully non-refundable rates, given the behaviour of the Non Refund bookings); a reminder or confirmation request, together with an invitation to personalise the stay, for medium-risk bookings; no action for low-risk guests. Overbooking, if used, should rely on the *aggregate* expected number of cancellations per date (the sum of the predicted probabilities) rather than on individual predictions.

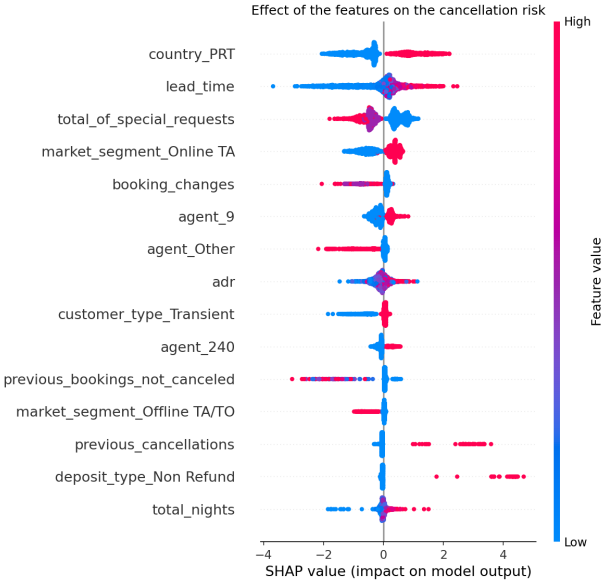


Figure 3. SHAP summary plot of XGBoost: each point is a booking; red points are high feature values, and points on the right increase the predicted cancellation risk.

TABLE 3. RISK SEGMENTATION OF THE TEST SET

	Low ($p < 0.4$)	Medium ($0.4 \leq p < 0.7$)	High ($p \geq 0.7$)
Share of bookings	56%	20%	24%
Actual cancellation rate	5.5%	34.8%	73.3%
Share of cancellations	11%	26%	63%
Avg. lead time (days)	63	79	119
Avg. special requests	0.77	0.96	0.32

4. Conclusion

This project applied the CRISP-DM methodology to predict hotel booking cancellations. The Data Understanding and Preparation phases proved decisive: removing duplicates and leakage lowered the apparent performance but produced a realistic model, and revealed that the Non Refund paradox was mostly an effect of duplicated group bookings. XGBoost was selected as the final model (PR-AUC 0.778, F1 0.701) and, with a threshold of 0.4, detects 89% of the cancellations. The main drivers are the guest's country, the lead time, the number of special requests and the booking channel, and a quarter of the bookings concentrates almost two thirds of the cancellations.

The main limitations are the performance drop observed on future bookings in the temporal validation, the data coming from only two Portuguese hotels in 2015–2017, a possible residual leakage in `booking_changes` and the

lack of cost data. Future work includes periodic retraining with more recent data, a cost-sensitive choice of the threshold based on real costs, and testing the proposed actions on a subset of bookings to measure their actual effect, since the relationships found are associations and not proven causal effects.

References

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